Loan Default Risk Analysis

You are a data analyst at a mid-size consumer lending company. Management is concerned about the rising default rate on personal loans and wants data-driven insights to improve the underwriting process. In this project you will explore borrower demographics, loan characteristics, and repayment outcomes to identify the key risk factors that predict loan defaults. This is a foundational credit risk analysis project that mirrors real work done at banks, fintech lenders, and credit unions.

Horizon Financial Group has issued over 600 personal loans across 2024 and 2025. The company has noticed that roughly 1 in 4 loans are defaulting, which is well above their target of 12%. The VP of Risk has asked your team to analyze the existing loan book and borrower data to answer key questions about what is driving defaults. Your analysis will directly inform changes to their credit scoring model and loan approval thresholds. You have been provided two datasets: one containing borrower profiles (demographics,

Resources

borrower_profiles.csv
CSV • 30.13 KB

loan_applications.csv
CSV • 49.13 KB

Questions to Answer

1. What is the overall default rate, and how does it break down by credit score range (e.g., 520–599, 600–649, 650–699, 700–749, 750+)? Which credit score bucket has the highest default rate?
2. Is there a relationship between a borrower's debt-to-income (DTI) ratio and the likelihood of defaulting? What DTI threshold would you recommend as a cutoff for loan approval?
3. Which loan purposes have the highest default rates, and does the average loan amount differ significantly between defaulted and non-defaulted loans?
4. How do employment status and years employed affect default risk? Are borrowers with less than 2 years of employment significantly more likely to default?

Recommended Tech Stack

Excel
SQL
Python - Pandas

What You Should Do

- Load both CSVs.
- Check for nulls, data types, and basic stats.
- Join the two tables on borrower_id.

2. Segment Analysis:

- Create credit score buckets and calculate default rates for each.
- Do the same for DTI ranges, employment status, and loan purpose.

3. Correlation Analysis:

- Identify which numeric variables (credit score, DTI, income, loan amount, interest rate) have the strongest correlation with defaults.

4. Visualization:

- Build charts showing default rate by credit score range, a scatter plot of DTI vs. default status, and a bar chart comparing loan purposes.

5. Recommendations:

- Write a brief summary recommending the top 3 risk factors and suggest specific thresholds (e.g., minimum credit score, maximum DTI) for the underwriting team.